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Sensing the World

3. *Prototypes*:
   - Prototype Roof mechanism to test before full integration
   - we will use a Breadboard and Arduino

4. *Performance*:
   the Roof succesfully opens and closes in response to the Presence of water or light.

5. *Usability*:
   can be used by people with low/bad mobility to still get the benifits of sun light and outdoors

6. *Safety*:
   Roof is in a stable position throughout whole operation

4. **Deliverables**
   1. *Reports*:
      Presentation on design and Functionality, Photos and vides of Prototype
   2. *Audits*: None
   3. *Outcomes*: we will create a Prototype System and design Presentation

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Project Requirements Document
Use the example from Dr. Valvano covered in class as a guide to complete this section to define your problem.

1. **Project Overview**
    1. *Objectives*:
    Create a prototype system to open and close a roof in response to two sensor inputs, an LDR and a moisture sensor. An LED will display the status of the roof
    2. *Process*:
    we will first prototype a mechanism for opening and closing the roof, once the prototype is functional we will wire it to the arduino and the rest of the system
        1. *Roles and Responsibilities*:
        max → Breadboarding & circuit design
        J.B. → Roof construction
        Josh → coding
        2. *Interactions with Existing Systems*:
        • Relies on an Arduino ESP32 microcontroller to control inputs & outputs
        • uses a photoresistor module to detect light
        3. *Terminology*:
        • Bread board - prototyping electronics board
        • SOC - system on chip
        • LED → Light Emitting diode
        • photo resistor → light sensor
        4. *Security*:
        • wires Hidden inside
        • sensors out of reach
3. **Function Description**
    1. *Functionality*:
    • The roof should be open and the LED should be on when there is light and no water
    • If there is water, or no light, roof should be closed.
    2. Scope: Hardware / Software
    • Prototype to showcase concept

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Project Worksheet

Objective: To break down your project objectives and specifics to make it more efficient to implement. This will help with practicing engineering design.

Submission:
🔥 Upload as a group to Gradescope once it has been signed off by course staff.

Team member names:
Joslin, Max Weiner, J. B. Volkert
Erica

Initial Planning

Consider the big picture and then answer the following questions.

1. What is the big system concept your prototype will emulate?
   Retractable roof

2. How does this solve a humanitarian problem?
   creates a comfortable living space

3. What aspects of the big system concept will be omitted in your prototype?
   the weight, and weather sealing

4. What sensors will be used in this prototype? How are they being used?
   our humidity sensor - to see if it's raining
   photoresistor / light sensor - to close at night
   button - to manually open and close